



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO KPL FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

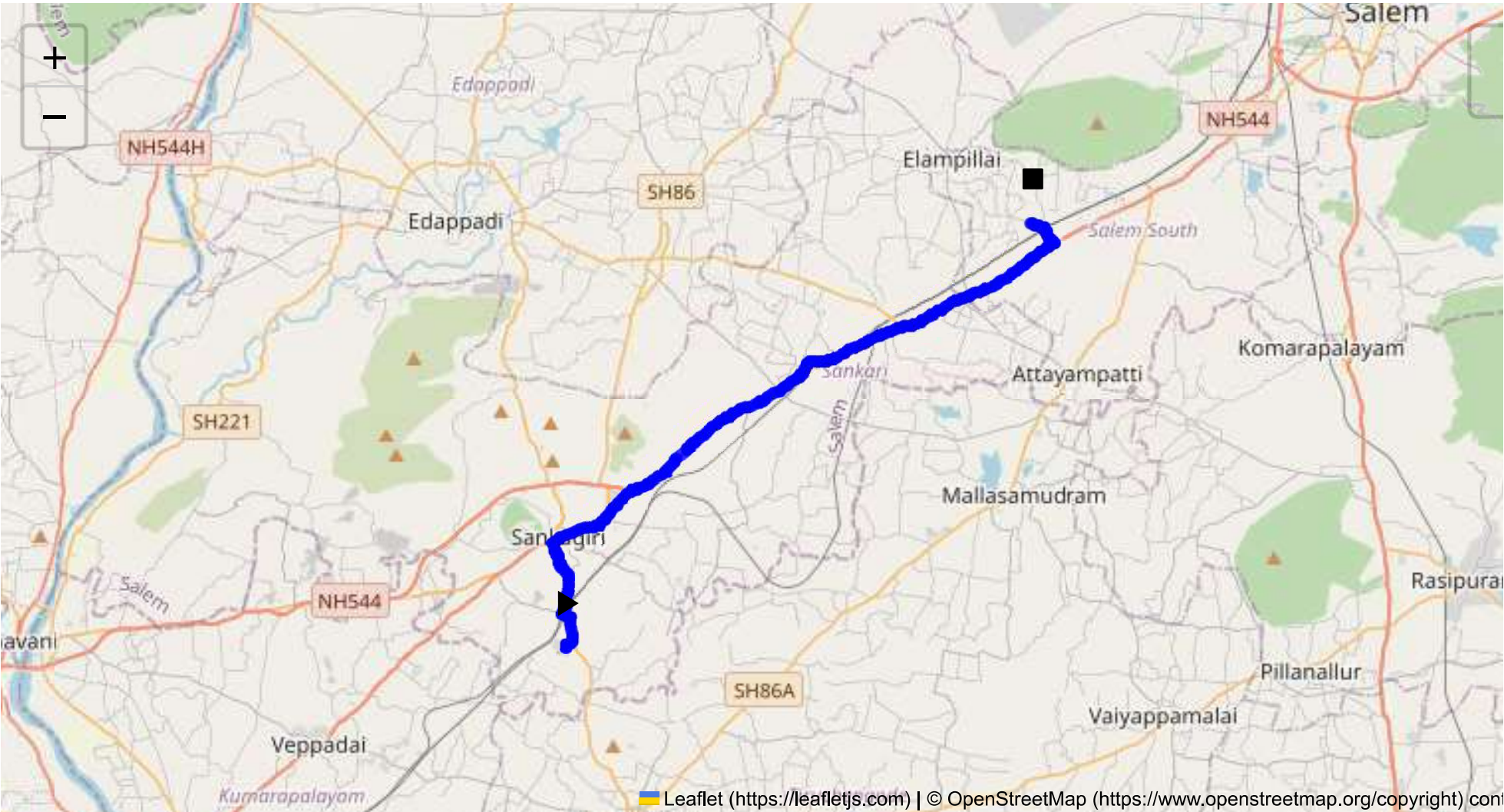
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 28.40 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.8 hours
Start: (11.4381, 77.8734)
End: (11.582969, 78.034683)



Welcome to the Journey Risk Management Study

Route Analysis for Safe Trucking from Sangagiri to Salem, Tamil Nadu

- Overview of the Route Map:** The journey from Sangagiri to Salem spans approximately 28.40 kilometers. This route predominantly follows NH 544, a major highway that connects southern districts. The route tends to be highly trafficked due to its strategic importance for commercial transportation, shortening travel times between the two locations.

2. **Typical Weather Conditions and Potential Weather-Related Hazards:** Tamil Nadu experiences a tropical climate. This stretch can experience heavy rainfall during the monsoon season (June to September) and high temperatures during the summer (March to May). Monsoon rains may lead to slippery roads and potential flooding, especially where drainage is poor.
3. **Traffic Patterns:** Traffic congestion is typically highest during morning (8 AM - 10 AM) and evening hours (6 PM - 8 PM) due to commuting patterns. Sections near urban areas, particularly nearer to Salem, regularly experience bottlenecks. Weekend traffic may also be high due to local market activities.
4. **Assessment of Road Quality and Infrastructure:** NH 544 is generally well-maintained, but there may be occasional potholes and construction work that can affect travel. Attention should be paid to any ongoing roadwork, which can cause unexpected detours or lane reductions.
5. **Suggestions for Alternative Routes for Emergencies:** In the event of significant delays or road closures, an alternative route via SH 86 may be considered, although this may increase travel time slightly due to lower speed limits and road quality concerns.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:** Transporting hazardous materials in Tamil Nadu requires adherence to specific state-imposed regulations, including proper labeling, documentation, and vehicle safety standards. Vehicles must follow designated time slots to minimize urban congestion impacts.
7. **Overview of Historical Incidents:** There have been occasional incidents involving heavy vehicles, usually attributed to brake failures or overloading. Hazardous material incidents are rare but underline the importance of maintenance and compliance with safety protocols.
8. **Environmental Considerations and Sensitive Areas:** The region near Salem includes some environmentally sensitive zones like agricultural lands. Special attention should be given to preventing leaks or spills, as they could significantly impact local agriculture and water sources.
9. **Analysis of Communication Coverage:** Mobile network coverage along NH 544 is generally robust; however, brief dead zones are possible in rural stretches with less commercial activity, impacting GPS and communication efficiency.
10. **Estimated Emergency Response Times:** Emergency services are more accessible closer to Salem, where the response time is approximately 20-30 minutes. In more rural stretches, estimate about 45 minutes to 1 hour for emergency services to arrive.
11. **Overall Summary of Risk Assessment:** The route is generally safe for the transportation of hazardous materials, with good road quality and infrastructure. Key risks include severe weather conditions, potential traffic congestion, and possible communication dead zones. Adherence to transport regulations and proactive planning, such as monitoring weather forecasts and traffic updates, can mitigate these risks effectively.

Trucking companies should ensure regular vehicle maintenance checks, especially the braking system given the terrain, and maintain clear lines of communication throughout the journey. Relief routes are limited but can be utilized in emergency scenarios to assure timely and safe deliveries.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	High	11.48369, 77.88781	15 KM/Hr
7	Turn	Medium	11.48370, 77.88787	30 KM/Hr
8	Turn	Medium	11.49149, 77.89601	30 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Blind Spot	Blind Spot	11.57645, 78.04266	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47407, 77.86839



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.48369, 77.88781



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48370, 77.88787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89601



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.57645, 78.04266